

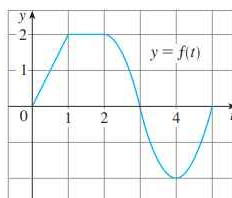
§4 The Fundamental Theorem of Calculus

* An equation of the form

$$g(x) = \int_a^x f(t) dt$$

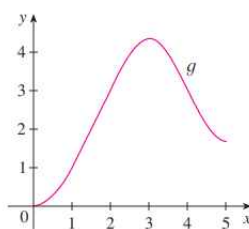
is a function with variable x where f is a continuous function on $[a, b]$.

Exam 1 If f is the function whose graph is shown as



and $g(x) = \int_0^x f(t) dt$. Sketch a rough graph of g .

(sol) $g(0) = 0$, $g(1) = 1$, $g(2) = 3$, $g(3) \approx 4.3$, $g(4) = 3$, $g(5) \approx 1.7$



Exam 2 If $g(x) = \int_a^x f(t) dt$, where $a = 1$ and $f(t) = t^2$, find a formula for $g(x)$ and calculate $g'(x)$.

(sol) $g(x) = \int_1^x t^2 dt = \frac{1}{3}t^3 \Big|_1^x = \frac{1}{3}x^3 - \frac{1}{3}$.

Then $g'(x) = x^2$

*** The Fundamental Theorem of Calculus, Part 1**

f : continuous on $[a, b]$

$$g(x) = \int_a^x f(t) dt, \quad a \leq x \leq b$$

$\Rightarrow g(x)$ is continuous on $[a, b]$ and differentiable, and $g'(x) = f(x)$.

(pf) Let $g(x) = \int_a^x f(t) dt$ and $x, x+h \in (a, b)$ with $h > 0$.

$$\begin{aligned} \Rightarrow g(x+h) - g(x) &= \int_a^{x+h} f(t) dt - \int_a^x f(t) dt \\ &= \int_a^x f(t) dt + \int_x^{x+h} f(t) dt - \int_a^x f(t) dt \\ &= \int_x^{x+h} f(t) dt \end{aligned}$$

$$\Rightarrow \frac{g(x+h) - g(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) dt$$

Since f is continuous on $[x, x+h]$, by the Extreme Value Theorem there are numbers u and v in $[x, x+h]$ such that $f(v) = m$ and $f(u) = M$, where m and M are the absolute minimum and maximum values of f on $[x, x+h]$.

$$\Rightarrow m \leq f(t) \leq M \quad \text{for } x \leq t \leq x+h$$

$$\Rightarrow mh \leq \int_x^{x+h} f(t) dt \leq Mh$$

$$\Rightarrow m \leq \frac{1}{h} \int_x^{x+h} f(t) dt \leq M$$

$$\Rightarrow f(v) \leq \frac{g(x+h) - g(x)}{h} \leq f(u)$$

Now, when $h \rightarrow 0^+$, $u \rightarrow x^+$ and $v \rightarrow x^+$.

Since $\lim_{h \rightarrow 0^+} f(v) = f(x) = \lim_{h \rightarrow 0^+} f(u)$, by the Squeeze Theorem,

$$\lim_{h \rightarrow 0^+} \frac{g(x+h) - g(x)}{h} = f(x)$$

By the similar manner, we get $\lim_{h \rightarrow 0^-} \frac{g(x+h) - g(x)}{h} = f(x)$.

$$\therefore g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = f(x).$$

Note $\frac{d}{dx} \int_a^x f(t) dt = f(x)$

Exam 3 Find the derivative of the function $g(x) = \int_0^x \sqrt{1+t^2} dt$.

(sol) By FTC, part 1, $g'(x) = \sqrt{1+x^2}$.

Exam 5 Find $\frac{d}{dx} \int_1^{x^4} \sec t dt$.

(sol) $\frac{d}{dx} \int_1^{x^4} \sec t dt = \frac{d}{dx} \int_1^u \sec t dt \quad (\text{Put } u = x^4)$
 $= \frac{d}{du} \int_1^u \sec t dt \cdot \frac{du}{dx}$
 $= \sec u \cdot 4x^3$
 $= 4x^3 \sec x^4$

(cf) $\frac{d}{dx} \int_{g_1(x)}^{g_2(x)} f(t) dt = \frac{d}{dx} \left[\int_{g_1(x)}^a f(t) dt + \int_a^{g_2(x)} f(t) dt \right]$
 $= \frac{d}{dx} \left[- \int_a^{g_1(x)} f(t) dt + \int_a^{g_2(x)} f(t) dt \right]$
 $= f(g_2(x)) g_2'(x) - f(g_1(x)) g_1'(x)$

* The Fundamental Theorem of Calculus

Suppose f is continuous on $[a, b]$.

1. If $g(x) = \int_a^x f(t) dt$, then $g'(x) = f(x)$.
2. $\int_a^b f(x) dx = F(b) - F(a)$, where F is any antiderivative of f , that is, $F' = f$.

* Average Value of a Function

The average value of finitely many numbers $y_1, y_2, \dots, y_n$ is

$$y_{ave} = \frac{y_1 + y_2 + \dots + y_n}{n}.$$

We want to find the average value of a function $y = f(x)$, $a \leq x \leq b$.

Let $\Delta x = \frac{b-a}{n}$ and $x_i = a + i \Delta x$ ($i = 0, 1, \dots, n$)

Let $x_i^* \in [x_{i-1}, x_i]$ ($i = 1, 2, \dots, n$) be the random points.

Then the average of the numbers $f(x_1^*), f(x_2^*), \dots, f(x_n^*)$ is

$$\begin{aligned} \frac{f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)}{n} &= \frac{f(x_1^*) + f(x_2^*) + \dots + f(x_n^*)}{\frac{b-a}{\Delta x}} \\ &= \frac{1}{b-a} \sum_{i=1}^n f(x_i^*) \Delta x \end{aligned}$$

Define the average value of f on the interval $[a, b]$ as

$$\begin{aligned} f_{ave} &= \lim_{n \rightarrow \infty} \frac{1}{b-a} \sum_{i=1}^n f(x_i^*) \Delta x \\ &= \frac{1}{b-a} \int_a^b f(x) dx \end{aligned}$$

Exam 6 Find the average value of the function $f(x) = 1 + x^2$ on the interval $[-1, 2]$.

(sol)

$$\begin{aligned} f_{ave} &= \frac{1}{2 - (-1)} \int_{-1}^2 (1 + x^2) dx \\ &= \frac{1}{3} \left[x + \frac{1}{3} x^3 \right]_{-1}^2 \\ &= 2 \end{aligned}$$

*** The Mean Value Theorem for Integrals**

f : continuous on $[a, b]$

$\Rightarrow$ There exists a number c in $[a, b]$ such that

$$f(c) = f_{ave} = \frac{1}{b-a} \int_a^b f(x) dx$$

that is

$$\int_a^b f(x) dx = f(c)(b-a)$$

(pf) Let $F(x) = \int_a^x f(t) dt$.

Since f is continuous on $[a, b]$, F is differentiable on $[a, b]$.

By the Mean Value Theorem, there exists a number c in $[a, b]$ such that

$$F'(c) = \frac{F(b) - F(a)}{b - a}.$$

Since $F'(x) = f(x)$, $F'(c) = f(c)$,

And, $F(b) = \int_a^b f(t) dt = \int_a^b f(x) dx$ and $F(a) = \int_a^a f(t) dt = 0$.

$$\therefore f(c) = \frac{\int_a^b f(x) dx}{b-a} = \frac{1}{b-a} \int_a^b f(x) dx.$$

Exam 7 Using the Mean Value Theorem for Integrals, find the number c in $[-1, 2]$ such that

$$\int_{-1}^2 (1+x^2) dx = f(c)[2 - (-1)]$$

(sol) Since $\int_{-1}^2 (1+x^2) dx = 6$, $6 = 3f(c) = 3(1+c^2)$.

$$\therefore c = \pm 1$$

